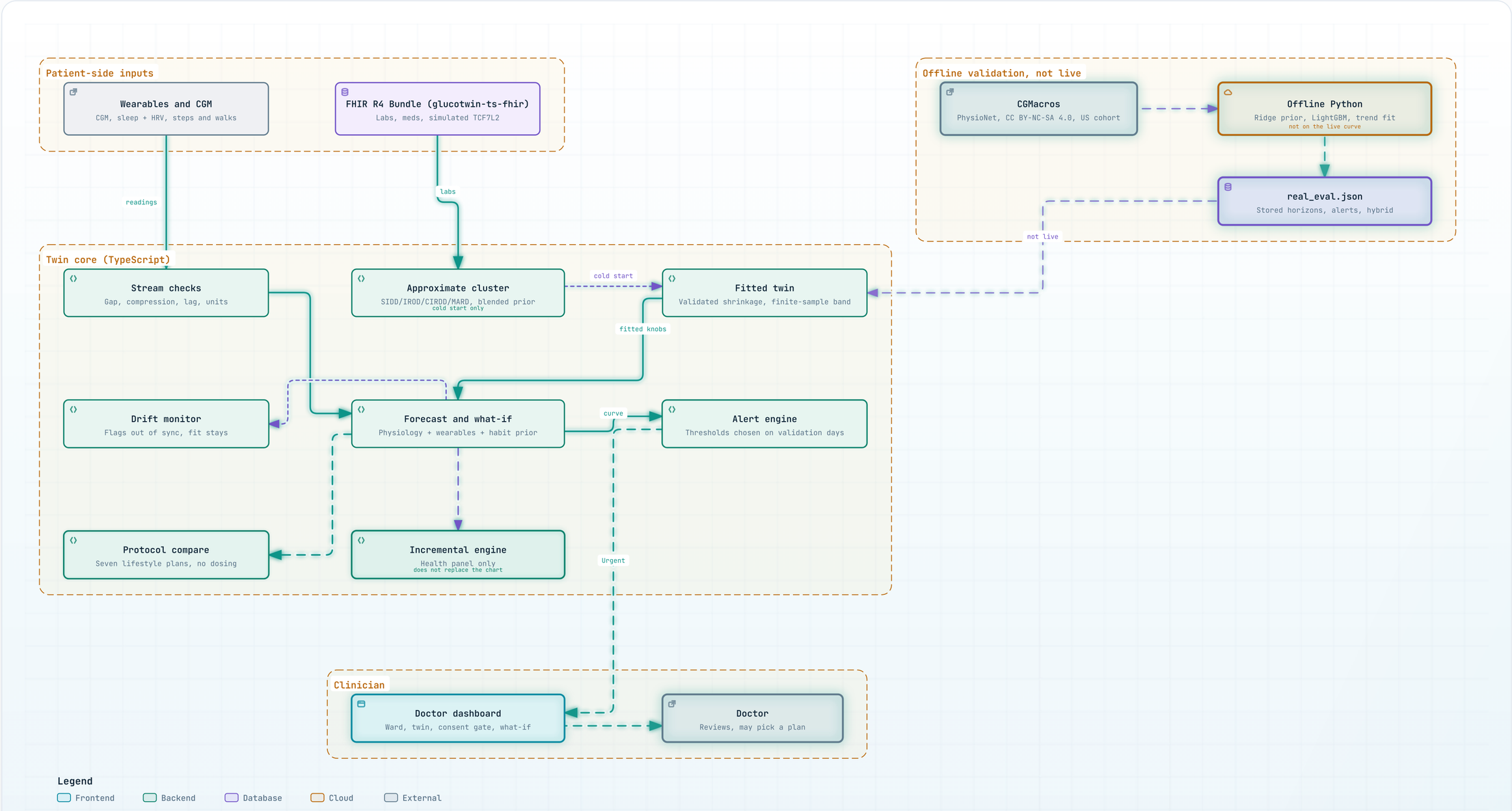
GlucoTwin — Type 2 diabetes digital twin

SIGNAL FLOW

Research prototype — not a medical device, not medical advice. All patient data is synthetic or open. Live forecast is physiology. LightGBM stays offline.



Two data streams

- Record: demographics, diagnoses, labs, and a simulated TCF7L2 allele count written as FHIR
- Wearables: CGM, heart rate, sleep stages and steps in an Apple Health-style file
- Per-stream consent gate in the dashboard: withdrawing the record drops the record prior and medicine list but keeps the personal CGM fit; withdrawing wearables drops sleep, HRV and walks; both widen the band, and alerts, plans and ranking follow

Twin loop that is actually on screen

- The live twin shrinks the personal fit toward a record + cluster prior; each is kept only when it clearly beats the group prior on validation days
- The chart calls forecastCurve with sleep, HRV and walk adjustments. The Kalman engine does not replace those knobs
- Spike risk adds the habitual next meal, so warnings can come up to 2 hours ahead
- What-if ranks meals, walks and seven lifestyle plans, including a weekly fasting day. No dosing

Trust and safety

- Only Urgent alerts interrupt; a Watch is upgraded in place when the curve worsens. Each alert has reasons
- Hazards, change bounds and intended use are in docs/HAZARDS.md and docs/REGULATORY.md
- One-click pre-visit clinical summary prints the twin, 7-day time in range, the chosen plan and urgent alerts

Not in this build

- No ABDM-certified import or export, and Synthea was not run
- No Web Worker on-device mode; the dashboard loads a committed snapshot of the fitted ward; reports/scale.json is a one-thread compute benchmark
- LightGBM is an offline CGMacros comparison, not a live residual
- Validated numbers are the stored CGMacros table plus the stress-test generator